

```
In [1]: import pandas as pd
```

Data Acquisition

<https://archive.ics.uci.edu/dataset/296/diabetes+130-us+hospitals+for+years+1999-2008>

```
In [3]: path = "../data/diabetic_data.csv"

df = pd.read_csv(path)

df.head()
```

```
Out[3]:
```

	encounter_id	patient_nbr	race	gender	age	weight	admission_type_id	dis
0	2278392	8222157	Caucasian	Female	[0-10)	?	6	
1	149190	55629189	Caucasian	Female	[10-20)	?	1	
2	64410	86047875	AfricanAmerican	Female	[20-30)	?	1	
3	500364	82442376	Caucasian	Male	[30-40)	?	1	
4	16680	42519267	Caucasian	Male	[40-50)	?	1	

5 rows × 50 columns

```
In [4]: df.shape
```

```
Out[4]: (101766, 50)
```

```
In [5]: df.info()
```

<class 'pandas.DataFrame'>

RangeIndex: 101766 entries, 0 to 101765

Data columns (total 50 columns):

#	Column	Non-Null Count	Dtype
0	encounter_id	101766 non-null	int64
1	patient_nbr	101766 non-null	int64
2	race	101766 non-null	str
3	gender	101766 non-null	str
4	age	101766 non-null	str
5	weight	101766 non-null	str
6	admission_type_id	101766 non-null	int64
7	discharge_disposition_id	101766 non-null	int64
8	admission_source_id	101766 non-null	int64
9	time_in_hospital	101766 non-null	int64
10	payer_code	101766 non-null	str
11	medical_specialty	101766 non-null	str
12	num_lab_procedures	101766 non-null	int64
13	num_procedures	101766 non-null	int64
14	num_medications	101766 non-null	int64
15	number_outpatient	101766 non-null	int64
16	number_emergency	101766 non-null	int64
17	number_inpatient	101766 non-null	int64
18	diag_1	101766 non-null	str
19	diag_2	101766 non-null	str
20	diag_3	101766 non-null	str
21	number_diagnoses	101766 non-null	int64
22	max_glu_serum	5346 non-null	str
23	A1Cresult	17018 non-null	str
24	metformin	101766 non-null	str
25	repaglinide	101766 non-null	str
26	nateglinide	101766 non-null	str
27	chlorpropamide	101766 non-null	str
28	glimepiride	101766 non-null	str
29	acetohexamide	101766 non-null	str
30	glipizide	101766 non-null	str
31	glyburide	101766 non-null	str
32	tolbutamide	101766 non-null	str
33	pioglitazone	101766 non-null	str
34	rosiglitazone	101766 non-null	str
35	acarbose	101766 non-null	str
36	miglitol	101766 non-null	str
37	troglitazone	101766 non-null	str
38	tolazamide	101766 non-null	str
39	examide	101766 non-null	str
40	citoglipton	101766 non-null	str
41	insulin	101766 non-null	str
42	glyburide-metformin	101766 non-null	str
43	glipizide-metformin	101766 non-null	str
44	glimepiride-pioglitazone	101766 non-null	str
45	metformin-rosiglitazone	101766 non-null	str
46	metformin-pioglitazone	101766 non-null	str
47	change	101766 non-null	str
48	diabetesMed	101766 non-null	str
49	readmitted	101766 non-null	str

```
dtypes: int64(13), str(37)
memory usage: 38.8 MB
```

```
In [6]: df.columns
```

```
Out[6]: Index(['encounter_id', 'patient_nbr', 'race', 'gender', 'age', 'weight',
              'admission_type_id', 'discharge_disposition_id', 'admission_source_id',
              'time_in_hospital', 'payer_code', 'medical_specialty',
              'num_lab_procedures', 'num_procedures', 'num_medications',
              'number_outpatient', 'number_emergency', 'number_inpatient', 'diag_1',
              'diag_2', 'diag_3', 'number_diagnoses', 'max_glu_serum', 'A1Cresult',
              'metformin', 'repaglinide', 'nateglinide', 'chlorpropamide',
              'glimepiride', 'acetohehexamide', 'glipizide', 'glyburide', 'tolbutamide',
              'pioglitazone', 'rosiglitazone', 'acarbose', 'miglitol', 'troglitazone',
              'tolazamide', 'examide', 'citoglipton', 'insulin',
              'glyburide-metformin', 'glipizide-metformin',
              'glimepiride-pioglitazone', 'metformin-rosiglitazone',
              'metformin-pioglitazone', 'change', 'diabetesMed', 'readmitted'],
             dtype='str')
```

Data Cleaning

- Clean target variable "readmitted"
- Convert target variable "readmitted" into boolean
- Missing Values
- Values with "?"

Convert Target Variable

```
In [9]: print(df["readmitted"].unique())
```

```
<StringArray>
['NO', '>30', '<30']
Length: 3, dtype: str
```

```
In [10]: # convert to string
df["readmitted"] = df["readmitted"].astype(str).str.strip()
```

```
In [11]: # convert boolean
df["readmitted"] = df["readmitted"].map({
    "<30": 1, # readmitted before 30 days
    ">30": 0, # readmitted after 30 days
    "NO": 0
})

df["readmitted"]
```

```
Out[11]: 0      0
         1      0
         2      0
         3      0
         4      0
         ..
        101761  0
        101762  0
        101763  0
        101764  0
        101765  0
        Name: readmitted, Length: 101766, dtype: int64
```

Values with '?'

```
In [13]: cols_with_question = df.columns[df.isin(['?']).any()].tolist()
         print(cols_with_question)
```

```
['race', 'weight', 'payer_code', 'medical_specialty', 'diag_1', 'diag_2', 'diag_3']
```

Replace with nan

```
In [15]: import numpy as np

         df.replace('?', np.nan, inplace=True)
```

Out[15]:

	encounter_id	patient_nbr	race	gender	age	weight	admission_type_i
0	2278392	8222157	Caucasian	Female	[0-10)	NaN	
1	149190	55629189	Caucasian	Female	[10-20)	NaN	
2	64410	86047875	AfricanAmerican	Female	[20-30)	NaN	
3	500364	82442376	Caucasian	Male	[30-40)	NaN	
4	16680	42519267	Caucasian	Male	[40-50)	NaN	
...	...	...	...	...	...	...	...
101761	443847548	100162476	AfricanAmerican	Male	[70-80)	NaN	
101762	443847782	74694222	AfricanAmerican	Female	[80-90)	NaN	
101763	443854148	41088789	Caucasian	Male	[70-80)	NaN	
101764	443857166	31693671	Caucasian	Female	[80-90)	NaN	
101765	443867222	175429310	Caucasian	Male	[70-80)	NaN	

101766 rows × 50 columns

Missing Values

```
In [17]: print( (df.isna().sum() / len(df) ) * 100)
```

encounter_id	0.000000
patient_nbr	0.000000
race	2.233555
gender	0.000000
age	0.000000
weight	96.858479
admission_type_id	0.000000
discharge_disposition_id	0.000000
admission_source_id	0.000000
time_in_hospital	0.000000
payer_code	39.557416
medical_specialty	49.082208
num_lab_procedures	0.000000
num_procedures	0.000000
num_medications	0.000000
number_outpatient	0.000000
number_emergency	0.000000
number_inpatient	0.000000
diag_1	0.020636
diag_2	0.351787
diag_3	1.398306
number_diagnoses	0.000000
max_glu_serum	94.746772
A1Cresult	83.277322
metformin	0.000000
repaglinide	0.000000
nateglinide	0.000000
chlorpropamide	0.000000
glimepiride	0.000000
acetohexamide	0.000000
glipizide	0.000000
glyburide	0.000000
tolbutamide	0.000000
pioglitazone	0.000000
rosiglitazone	0.000000
acarbose	0.000000
miglitol	0.000000
trogliatone	0.000000
tolazamide	0.000000
examide	0.000000
citoglipton	0.000000
insulin	0.000000
glyburide-metformin	0.000000
glipizide-metformin	0.000000
glimepiride-pioglitazone	0.000000
metformin-rosiglitazone	0.000000
metformin-pioglitazone	0.000000
change	0.000000
diabetesMed	0.000000
readmitted	0.000000
dtype: float64	

Features will be considered during transformation/pipelining. Dropping payer_code as its a insurance code.

```
In [19]: df.drop('payer_code', axis=1, inplace=True)
```

```
In [20]: print( (df.isna().sum() / len(df) ) * 100)
```

encounter_id	0.000000
patient_nbr	0.000000
race	2.233555
gender	0.000000
age	0.000000
weight	96.858479
admission_type_id	0.000000
discharge_disposition_id	0.000000
admission_source_id	0.000000
time_in_hospital	0.000000
medical_specialty	49.082208
num_lab_procedures	0.000000
num_procedures	0.000000
num_medications	0.000000
number_outpatient	0.000000
number_emergency	0.000000
number_inpatient	0.000000
diag_1	0.020636
diag_2	0.351787
diag_3	1.398306
number_diagnoses	0.000000
max_glu_serum	94.746772
A1Cresult	83.277322
metformin	0.000000
repaglinide	0.000000
nateglinide	0.000000
chlorpropamide	0.000000
glimepiride	0.000000
acetohexamide	0.000000
glipizide	0.000000
glyburide	0.000000
tolbutamide	0.000000
pioglitazone	0.000000
rosiglitazone	0.000000
acarbose	0.000000
miglitol	0.000000
troglitazone	0.000000
tolazamide	0.000000
examide	0.000000
citoglipton	0.000000
insulin	0.000000
glyburide-metformin	0.000000
glipizide-metformin	0.000000
glimepiride-pioglitazone	0.000000
metformin-rosiglitazone	0.000000
metformin-pioglitazone	0.000000
change	0.000000
diabetesMed	0.000000
readmitted	0.000000

dtype: float64

Convert Categorical Data

```
In [22]: df_cat = df.select_dtypes(include='object')
df_cat.head(5)
```

C:\Users\Matt\AppData\Local\Temp\ipykernel_23992\1324786334.py:1: Pandas4Warning: For backward compatibility, 'str' dtypes are included by select_dtypes when 'object' dtype is specified. This behavior is deprecated and will be removed in a future version. Explicitly pass 'str' to 'include' to select them, or to 'exclude' to remove them and silence this warning.

See https://pandas.pydata.org/docs/user_guide/migration-3-strings.html#string-migration-select-dtypes for details on how to write code that works with pandas 2 and 3.

```
df_cat = df.select_dtypes(include='object')
```

Out[22]:

	race	gender	age	weight	medical_specialty	diag_1	diag_2	diag_3	max_g
0	Caucasian	Female	[0-10)	NaN	Pediatrics-Endocrinology	250.83	NaN	NaN	
1	Caucasian	Female	[10-20)	NaN	NaN	276	250.01	255	
2	AfricanAmerican	Female	[20-30)	NaN	NaN	648	250	V27	
3	Caucasian	Male	[30-40)	NaN	NaN	8	250.43	403	
4	Caucasian	Male	[40-50)	NaN	NaN	197	157	250	

5 rows × 35 columns

Correlations

```
In [24]: corr = df.corr(numeric_only=True)["readmitted"].sort_values(ascending=False).head(1)
print(corr)
```

```
readmitted      1.000000
number_inpatient 0.165147
number_emergency 0.060747
discharge_disposition_id 0.050560
number_diagnoses 0.049524
time_in_hospital 0.044199
num_medications  0.038432
num_lab_procedures 0.020364
number_outpatient 0.018893
patient_nbr      0.007893
Name: readmitted, dtype: float64
```

```
In [25]: import matplotlib.pyplot as plt
```

```
# select numeric columns
```



```

numeric_df = df.select_dtypes(include="number")

# correlation matrix
corr = numeric_df.corr()

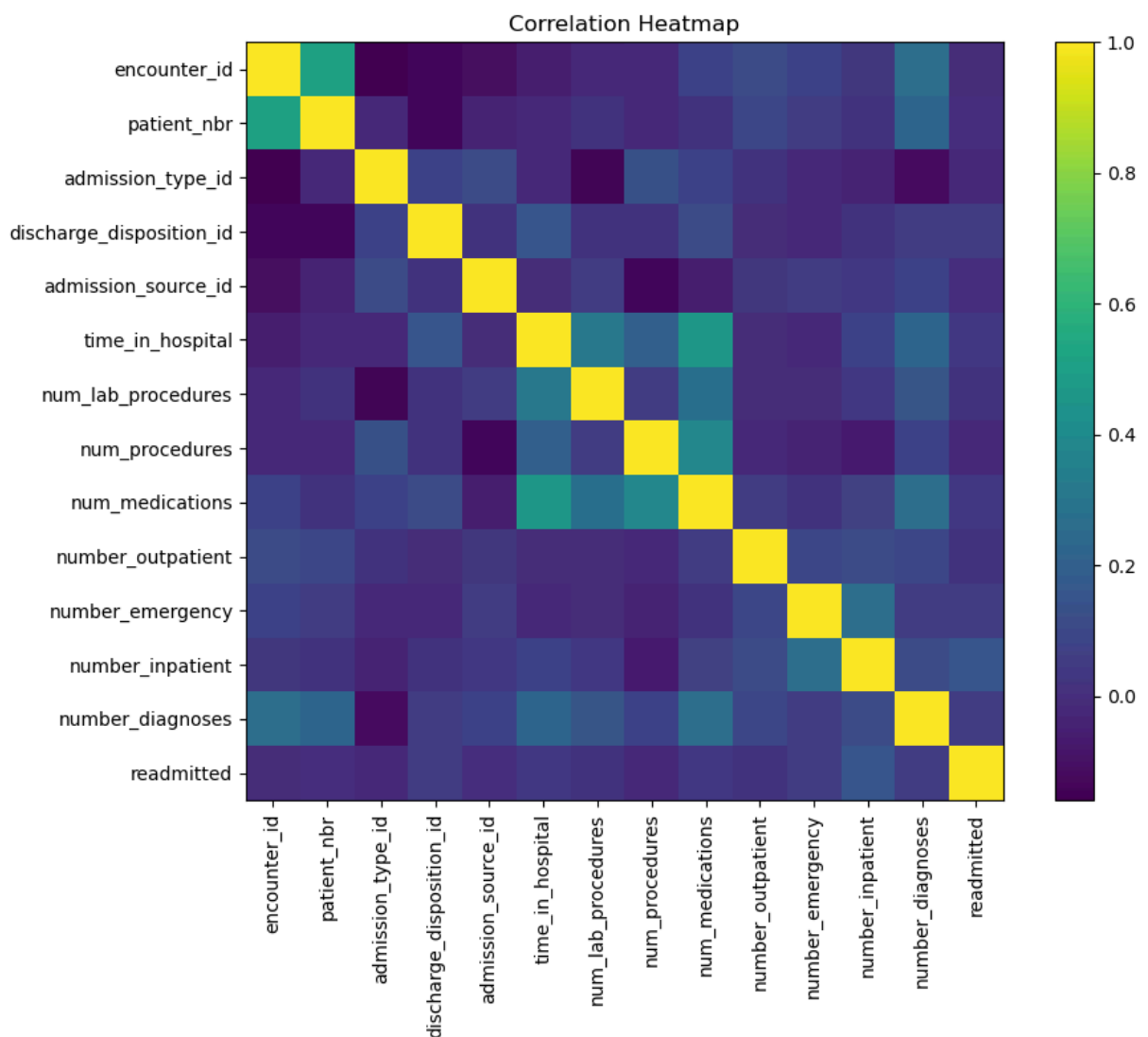
plt.figure(figsize=(10,8))
plt.imshow(corr)

plt.colorbar()
plt.xticks(range(len(corr.columns)), corr.columns, rotation=90)
plt.yticks(range(len(corr.columns)), corr.columns)

plt.title("Correlation Heatmap")

plt.tight_layout()
plt.show()

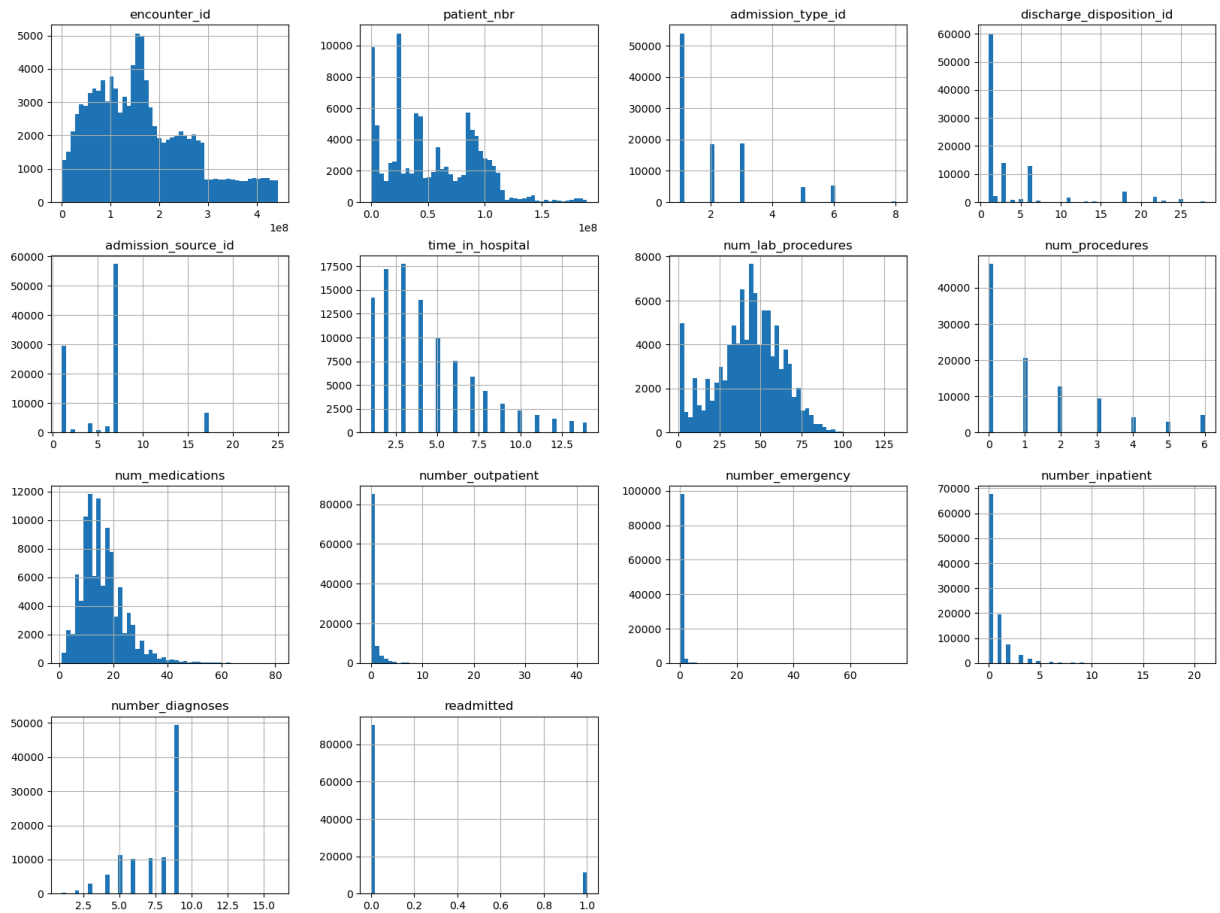
```



Distributions

```
In [27]: import matplotlib.pyplot as plt
```

```
df.hist(bins=50, figsize=(20,15))
plt.show()
```

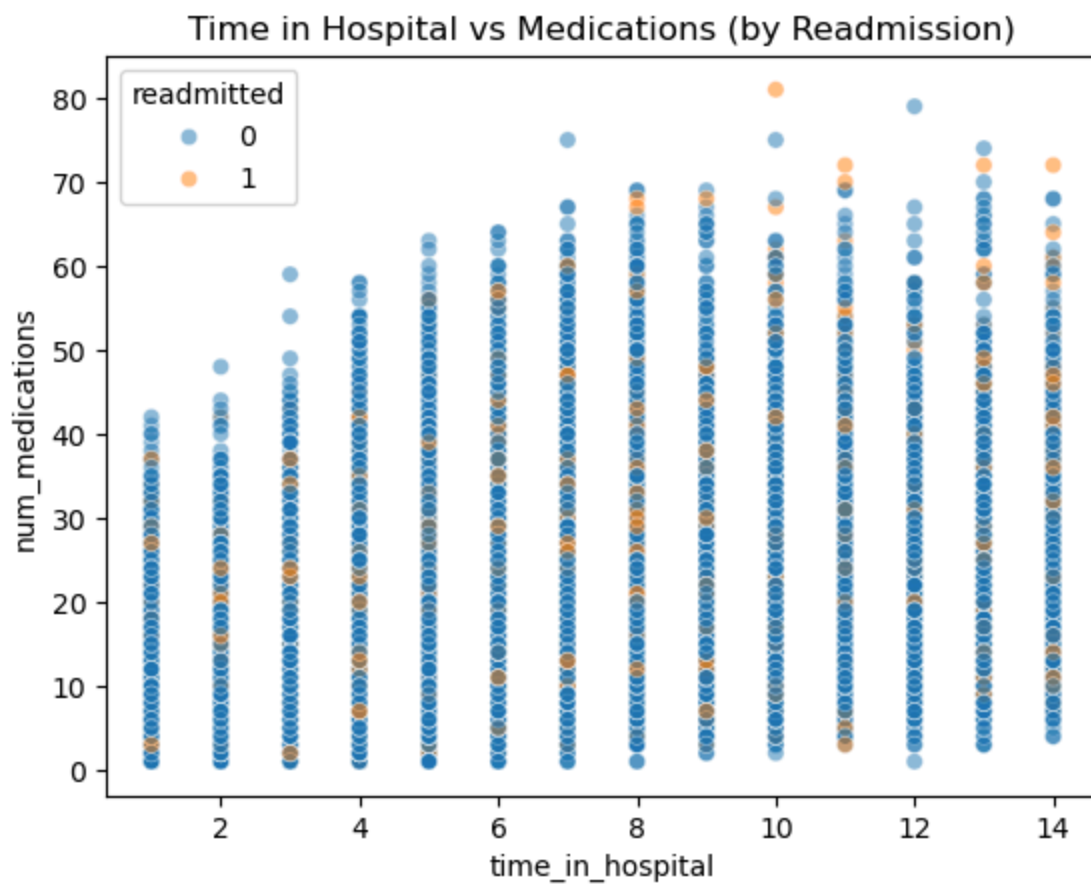


EDA

```
In [29]: import seaborn as sns

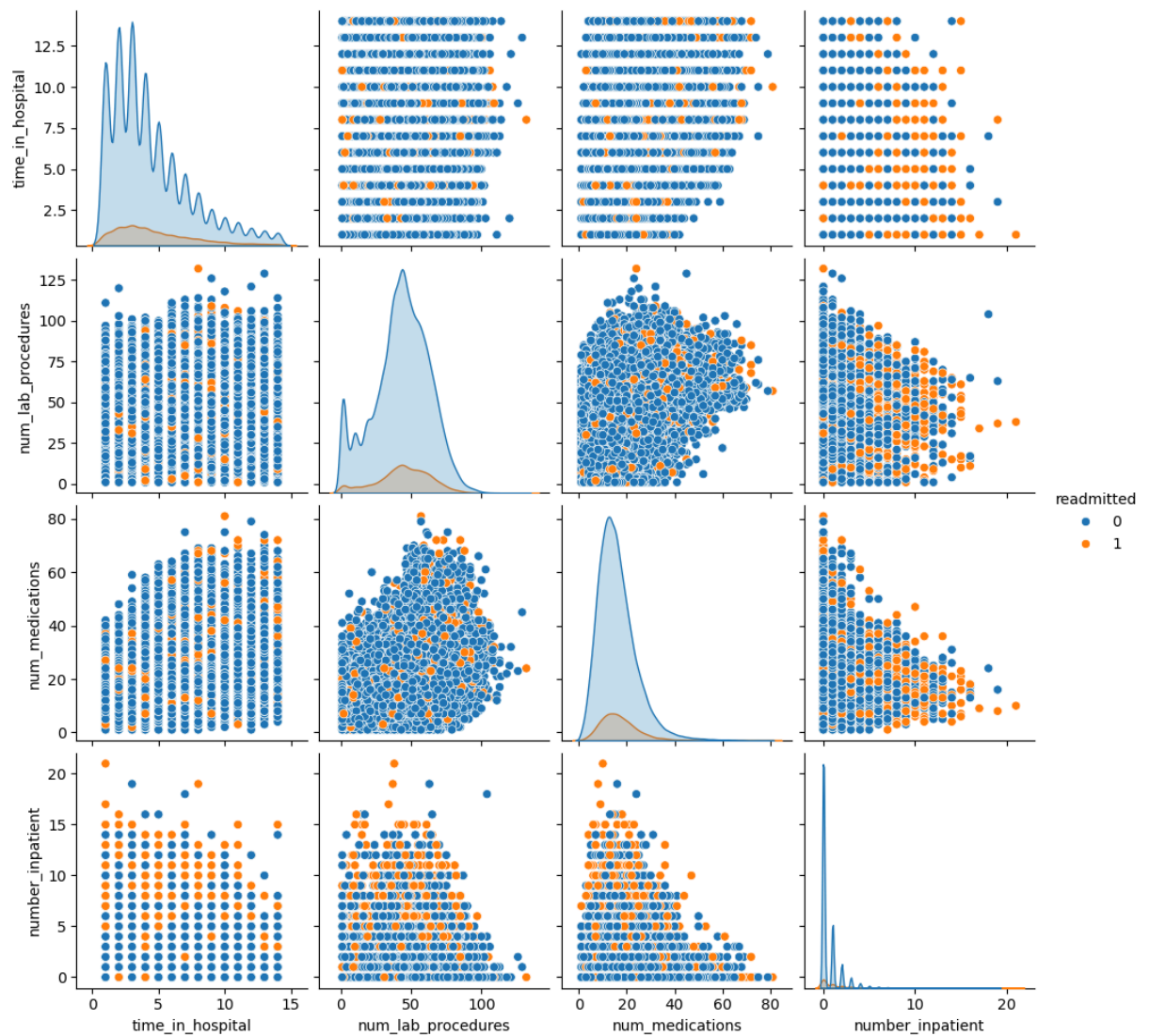
sns.scatterplot(
    data=df,
    x='time_in_hospital',
    y='num_medications',
    hue='readmitted',
    alpha=0.5
)

plt.title('Time in Hospital vs Medications (by Readmission)')
plt.show()
```



```
In [30]: cols = ['time_in_hospital', 'num_lab_procedures', 'num_medications', 'number_inpatient']  
sns.pairplot(df[cols + ['readmitted']], hue='readmitted')
```

```
Out[30]: <seaborn.axisgrid.PairGrid at 0x1275943d2b0>
```

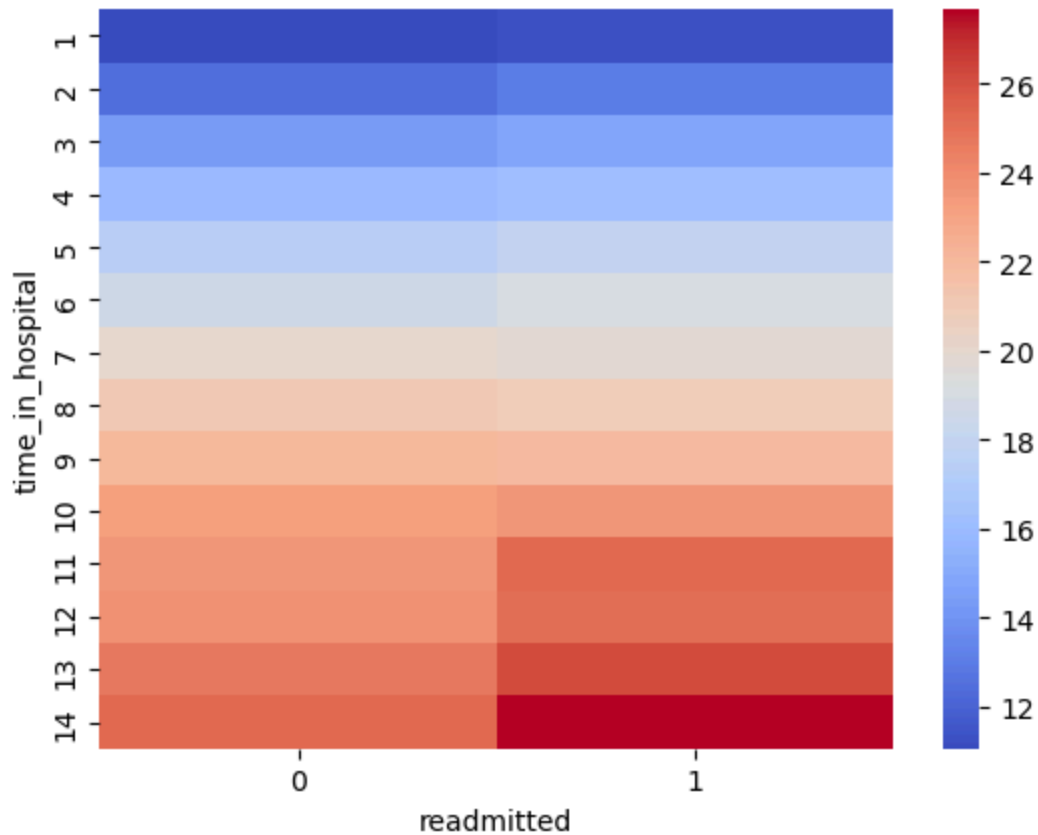


Time in hospital vs readmitted

```
In [32]: pivot = df.pivot_table(
    values='num_medications',
    index='time_in_hospital',
    columns='readmitted',
    aggfunc='mean'
)

sns.heatmap(pivot, cmap='coolwarm')
```

Out[32]: <Axes: xlabel='readmitted', ylabel='time_in_hospital'>



The heatmap shows a positive relationship between time_in_hospital and number of medications- as time in hospital increases the number of medications also increase. This is prominent between 10-14 days of readmitted patients as the number of medications is much greater than non readmitted with the color being much darker red.

```
In [34]: df_corr = df.copy()
```

```
In [35]: df_corr.shape
```

```
Out[35]: (101766, 49)
```

```
In [36]: df_corr["severity"] = df["time_in_hospital"] * df["num_medications"]
```

```
In [37]: df_corr['total_visits'] = (
    df_corr['number_inpatient'] +
    df_corr['number_outpatient'] +
    df_corr['number_emergency']
)
```

```
In [38]: df_corr['com'] = df["number_diagnoses"] + df["num_medications"] + df["num_medication"]
```

```
In [39]: corr = df_corr.corr(numeric_only=True)["readmitted"].sort_values(ascending=False)
print(corr)
```

```
readmitted          1.000000
number_inpatient    0.165147
total_visits        0.126114
number_emergency    0.060747
discharge_disposition_id  0.050560
number_diagnoses    0.049524
time_in_hospital    0.044199
com                 0.042722
severity            0.041994
num_medications     0.038432
num_lab_procedures  0.020364
number_outpatient   0.018893
patient_nbr         0.007893
admission_source_id  0.005820
encounter_id        -0.008519
admission_type_id   -0.011651
num_procedures      -0.012227
Name: readmitted, dtype: float64
```

The correlation graph shows that groups the number of visits together present a potential strong feature when predicting readmitted patients.

Saving

```
In [42]: df.to_csv("../data/df_diabetic.csv", index=False)
```